📘 Project Overview (What Exactly You Are Building)

FinPulse is an advanced web-based financial intelligence platform that:

- Collects financial transaction & market data
- Processes data using ETL pipelines
- Stores structured data in PostgreSQL
- Applies Machine Learning models for:
    - Revenue forecasting
    - Risk detection
    - Anomaly detection
- Secures financial records using:
    - SHA-256 blockchain hashing
    - JWT authentication
    - Encrypted storage
- Displays insights through a modern React dashboard

This is a **FinTech + AI + Cybersecurity + Blockchain** system.

🧱 TECHNOLOGY STACK (FINAL)

Backend

- Python
- FastAPI
- PySpark

Frontend

- React.js
- Tailwind CSS

Database

- PostgreSQL

Machine Learning

- Scikit-learn

- XGBoost
- Isolation Forest

Blockchain

- Private Blockchain (Python)
- SHA-256 hashing

Security

- JWT Authentication
- bcrypt password hashing
- AES encryption (optional advanced)

📅 FULL DEVELOPMENT ROADMAP (60 DAYS – ADVANCED LEVEL)

🟢 PHASE 1 – FOUNDATION (Day 1–7)

Day 1 – Final Architecture Design

- Draw system architecture
- Design data flow diagram
- Plan database schema
- Assign team roles

Day 2 – Environment Setup

- Install Python 3.10+
- Install Node.js
- Install PostgreSQL
- Setup virtual environment
- Install required libraries

Day 3 – Backend Initialization

- Create FastAPI project
- Setup folder structure
- Test basic API endpoint

Day 4 – Database Setup

- Create PostgreSQL database
- Design tables:
  - users
  - financial_data
  - predictions
  - blockchain_log

Day 5 – Authentication System

- Register API
- Login API
- JWT token generation
- Password hashing with bcrypt

Day 6 – Role-Based Access

- Admin
- Analyst
- Auditor

Day 7 – Connect Backend + Database

- Test CRUD operations

✅ Phase 1 Output: Secure backend foundation

🟡 **PHASE 2 – DATA ENGINEERING (Day 8–18)**

Day 8–9 – File Upload API

- CSV upload endpoint
- Validate file type

Day 10–12 – ETL Pipeline

- Clean column names
- Handle missing values
- Convert numeric fields

- Normalize data

Day 13–15 – PySpark Integration

- Setup Spark session
- Process large datasets
- Simulate distributed processing

Day 16–18 – KPI Engine

Generate:

- Total revenue
- Average transaction
- Max transaction
- Monthly growth rate

✅ Phase 2 Output: Working ETL + Data Processing Engine

🔵 **PHASE 3 – MACHINE LEARNING (Day 19–30)**

Day 19–22 – Revenue Forecasting

- Feature engineering
- Train XGBoost / Regression model
- Predict next month revenue
- Save prediction in DB

Day 23–25 – Anomaly Detection

- Isolation Forest
- Flag suspicious transactions
- Store risk level

Day 26–27 – Risk Classification

- Logistic Regression or XGBoost classifier
- Classify:
    - Low risk
    - Medium risk

    - High risk

Day 28–30 – ML API Integration

- Connect models to FastAPI
- Create:
    - /forecast endpoint
    - /detect-risk endpoint

✅ Phase 3 Output: AI system working

🟣 **PHASE 4 – BLOCKCHAIN SECURITY (Day 31–40)**

Day 31–33 – SHA-256 Hashing

- Hash financial dataset
- Hash ML results

Day 34–36 – Private Blockchain

Block structure:

- Index
- Timestamp
- Data
- Previous hash
- Current hash

Day 37–38 – Integrity Verification

- Recalculate hash
- Compare blockchain chain
- Show:
    - Valid
    - Tampered

Day 39–40 – Store Blockchain Log in DB

✅ Phase 4 Output: Tamper-proof validation system

🟠 **PHASE 5 – FRONTEND DEVELOPMENT (Day 41–52)**

Day 41–42 – React Setup

- Initialize React project
- Setup routing

Day 43–45 – Authentication UI

- Login page
- Registration page
- JWT storage

Day 46–48 – Dashboard UI

Display:

- KPIs
- Revenue forecast graph
- Risk summary
- Blockchain validation status

Day 49–50 – Charts

- Line chart (revenue trend)
- Pie chart (risk distribution)
- Bar chart (monthly comparison)

Day 51–52 – Connect React to FastAPI

✅ Phase 5 Output: Professional dashboard

🔴 **PHASE 6 – SECURITY + FINAL TESTING (Day 53–60)**

Day 53–54 – Encryption Layer

- Encrypt sensitive financial fields

Day 55–56 – Performance Testing

- Test large dataset
- Measure response time

Day 57–58 – Bug Fixing

- Fix UI issues
- Fix API errors

Day 59 – Deployment

- Docker container
- Run locally / Cloud deploy

Day 60 – Documentation & PPT

- Architecture diagram
- ER diagram
- Screenshots
- Report
- Viva preparation

🎯 **Final Output**

You will have:

✓ Advanced Web Application
✓ AI Forecasting Engine
✓ Anomaly Detection
✓ Blockchain Validation
✓ Cybersecurity Layer
✓ Modern React Dashboard
✓ Industry-Level Architecture